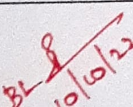
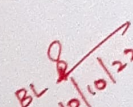
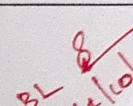
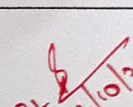
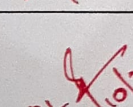
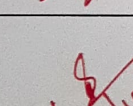
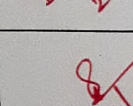
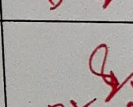
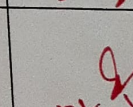
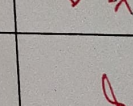
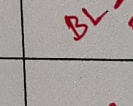


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4	14/10/22	SELECTION SORT	27	10	BL  14/10/22
5	21/10/22	BUBBLE SORT	37	10	BL  21/10/22
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20					

blotted!

AIM:

To write a C program to search a particular element from the given set of elements using linear search.

ALGORITHM:

Variables used:

$K \leftarrow$ array to hold elements

$N \leftarrow$ Total no. of elements

$X \leftarrow$ Element to be searched

Function LINEAR SEARCH (K, N, X)

1. [Initialize search]

$I \leftarrow 1$

$K[N+1] \leftarrow X$

2. [Search the vector]

Repeat while $K[I] \neq X$

$I \leftarrow I+1$

3. [Successful Search]

If $I = N+1$

Then write ("UNSUCCESSFUL SEARCH")

Return (0).

Else write ("SUCCESSFUL SEARCH")

Return (1)

LINEAR SEARCH

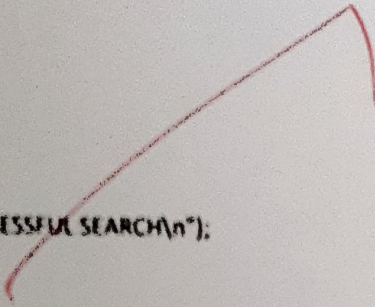
Source code:

```
#include<stdio.h> int
main()
{
    int a[100],n,i,k,flag=0;
    printf("\n\nLINEAR SEARCH\n\n");
    printf("Enter the Number of elements");
    scanf("%d",&n); for(i=0;i<n;i++)
    {
        scanf("%d",&a[i]);
    }
    printf("Enter the element to search\n");
    scanf("%d",&k);
    i=0;
    while((flag!=1)&&(k>=a[i]))
    {
        if(k==a[i])
        {
            flag=1;
            printf("\nSUCCESSFUL SEARCH\n");
        }
    }

    else
    {
        i=i+1; }
    if(i>n)
```


--	--	--

```
{  
    break;  
}  
}  
if(flag==0)  
{  
    printf("\nUNSUCCESSFUL SEARCH\n");  
}  
return 0;  
}
```



OUTPUT

```
LINEAR SEARCH
Enter the Number of elements5
23
29
37
38
100
Enter the element to search
100

SUCCESSFUL SEARCH

Process returned 0 (0x0)   execution time : 32.047 s
Press any key to continue.
```

RESULT:

Thus the linear search program is executed and verified successfully.

Handwritten signature

EXP NO: 2

BINARY SEARCH

DATE:

10/10/22

AIM:

To write a C program to search a particular element from the given set of elements using Binary search.

ALGORITHM:

Variables used:

$k \leftarrow$ Vector to hold the elements

$x \leftarrow$ Target element to be searched

$LOW \leftarrow$ Points to the lower bound of the vector

$HIGH \leftarrow$ Points to the upper bound of the vector

$MIDDLE \leftarrow$ Points to the middle element in the vector.

1. [Initialize]

$LOW \leftarrow 1$

$HIGH \leftarrow N$

2. [Perform search]

Repeat the step 4 while

$LOW \leq HIGH$

3. [Obtain Index of midpoint interval]

$MIDDLE \leftarrow (LOW + HIGH) / 2$

4. [COMPARE]

If $x < k[MIDDLE]$

Then $HIGH \leftarrow MIDDLE - 1$

Else if $x > k[MIDDLE]$

Then $LOW \leftarrow MIDDLE + 1$

Else write ('SUCCESSFUL SEARCH')

Return (MIDDLE)

5. [Unsuccessful search]

Write ('UNSUCCESSFUL SEARCH')

Return (0).

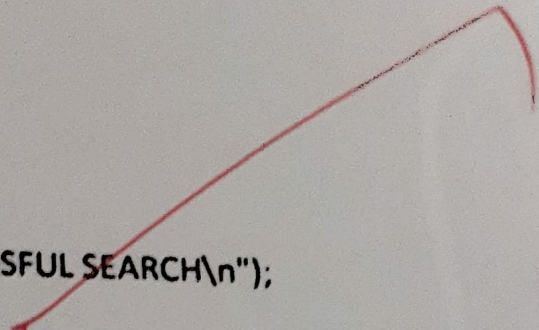
BINARY SEARCH

Source code:

```
#include<stdio.h> int
main()
{
    int a[100],n,i,l,u,mid,flag=0,k;
    printf("\n\n BINARY SEARCH\n\n");
    printf("Enter the Number of elements :");
    scanf("%d",&n); printf("Enter the
elements :\n"); for(i=0;i<n;i++)
    {
        scanf("%d",&a[i]);
    }
    printf("Enter the element to search\n");
    scanf("%d",&k);
    l=0; u=n;
    while((flag!=1)&&(l<u))
    {
        mid=(l+u)/2;
        if(k==a[mid]) { flag=1;
        printf("\nSUCCESSFUL SEARCH\n");
        }
        if(k<a[mid])
        {
            u=mid-1;
        }
    }
```


--	--	--

```
else
{
    l=mid+1;
}
}
if(flag==0)
{
    printf("\nUNSUCCESSFUL SEARCH\n");
}
return 0;
}
```



BINARY SEARCH

Enter the Number of elements5

56

52

78

37

38

Enter the element to search

37

UNSUCCESSFUL SEARCH

Process returned 0 (0x0) execution time : 24.188 s
Press any key to continue.

RESULT:

Thus the binary search program executed and verified successfully.

✓